

Olympiad Test Paper — Science (Class 7) — Paper 5

Chapter 4: Heat

Paper 5 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	4 — Heat
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Hardest tier: weighted-mixing where the answer lands nearer the larger share (never the plain midpoint); equilibrium reasoning with no temperature difference; staged scale-division reads where small marks are halves, fifths or coarse whole degrees (never automatically tenths); rate-of-change extrapolations; multi-mode chains where you must name the mode at each step; emitter-and-absorber duality of dull-dark vs shiny surfaces; breeze reversals tied to which surface gains or loses heat faster; and 'feels colder is colder' versus what a thermometer would actually read. Do not write on this paper — mark answers on your answer sheet.

1. 100 g of water at 90 °C is mixed with 300 g of water at 10 °C in a perfectly insulated jar. Using only the idea that equal amounts share the gap equally and that the larger share dominates, the final temperature is closest to:

- (A) 50 °C, the plain midpoint of 10 and 90, because mixing always gives the average of the two temperatures
- (B) 70 °C, much nearer the hot value, because hotter water always controls the final temperature
- (C) 100 °C, because the two temperatures (90 and 10) add to make the total
- (D) 30 °C, because the 300 g of cold water is three times the hot share, so the mix settles much nearer the cold value than the midpoint

2. A 50 g block of metal at 100 °C is dropped into 50 g of water at 20 °C in an insulated cup. Knowing that water needs far more heat than most metals to change its temperature by one degree, the final shared temperature will be:

- (A) exactly 60 °C, the midpoint of 100 and 20, since the two masses are equal
- (B) much closer to 100 °C, because the metal is far hotter and so controls the result
- (C) much closer to 20 °C than to 60 °C, because the water resists temperature change strongly, so a small temperature rise in the water absorbs the metal's heat
- (D) exactly 20 °C, because water never lets its temperature change once it is poured

3. Two identical sealed cans, one holding 1 litre of water and one holding 1 litre of cooking oil, are both heated on the same flame for the same time and both reach the same temperature. A student claims they each absorbed the same amount of heat. The soundest judgement is:

- (A) they did not necessarily absorb equal heat — reaching the same temperature only tells you their final hotness is equal, not the heat each took in, because water and oil need different amounts of heat to warm by the same number of degrees
- (B) they absorbed exactly equal heat, because reaching the same temperature always means equal heat content
- (C) the oil absorbed more heat, because oil is always hotter than water at the same flame
- (D) neither absorbed any heat, because heat and temperature are unrelated quantities

4. A large tank of water at 25 °C and a tiny metal coin at 25 °C sit in the same warm room at 25 °C. A student says heat must flow from the big tank to the small coin because the tank holds far more total heat. The correct view is:

- (A) heat flows from the tank to the coin, because the tank holds more total heat and must release some
- (B) heat flows from the coin to the tank, because small objects always lose heat to large ones
- (C) heat flows from the room to the coin, because rooms always warm small metal objects
- (D) no net heat flows in either direction, because heat moves only when there is a temperature difference, and all three are already at the same temperature

5. A cup of hot tea at 70 °C is poured into a thick, cold ceramic mug that was at 20 °C. Before the tea reaches your lips it has cooled noticeably even though the room is warm. The main reason the tea cooled so quickly at first is:

- (A) the cold mug pushed cold into the tea, lowering the tea's temperature
- (B) heat flowed from the hot tea into the cold mug until tea and mug neared a common temperature, and the cold mug soaked up a lot of that heat
- (C) the warm room cooled the tea by sending warm air into it
- (D) the tea created less heat once it was poured, so its temperature dropped

6. Equal amounts of the same water are warmed: sample P from 10 °C to 25 °C, and sample Q from 50 °C to 60 °C. Which sample took in MORE heat, and why?

- (A) sample Q took in more heat, because it ended at the higher temperature of 60 °C
- (B) sample P took in more heat, because it rose by 15 degrees while Q rose by only 10, and equal water needs more heat for a larger rise
- (C) both took in the same heat, because they are equal amounts of the same water
- (D) neither took in heat, because the starting temperatures were different

7. A hot iron ball at 200 °C is dropped into a small bowl of cold water; the water warms a little and the ball cools a lot. Why does the ball's temperature fall by far more degrees than the water's temperature rises?

- (A) the same heat leaves the ball and enters the water, but warming that water by one degree takes much more heat than cooling the ball by one degree, so the ball changes by many more degrees
- (B) the ball loses more heat than the water gains, because some heat simply disappears in the transfer
- (C) the water rises by exactly the number of degrees the ball falls, since heat is shared equally
- (D) the ball cools more because hot metal always loses all of its temperature instantly

8. A technician must (i) record the temperature of a freezing brine at about -6 °C, (ii) check a baby's bath water is exactly 38.4 °C, and (iii) carry that bath reading across the room to log it. Only a clinical thermometer (35–42 °C, with a kink) and a laboratory thermometer (-10 °C to 110 °C, no kink) are available. The correct plan is:

- (A) use the laboratory thermometer for the -6 °C brine, and the clinical thermometer for the 38.4 °C bath that must be carried away, because its kink holds the reading
- (B) use the clinical thermometer for all three, because it reads body-warm temperatures in Celsius
- (C) use the laboratory thermometer for all three, since its wide range covers every value
- (D) use the clinical thermometer for the -6 °C brine and the laboratory thermometer for the carried bath reading

9. A nurse reads 40.1 °C on a clinical thermometer from a feverish child but forgets to shake it down. She then places the same thermometer under a second child whose true temperature is 38.5 °C. What does she read, and why?

- (A) about 40.1 °C, because the kink holds the thread at the earlier higher value and a 38.5 °C body is not warm enough to push it any higher
- (B) about 38.5 °C, because the thermometer resets to each new patient automatically
- (C) about 78.6 °C, because the two readings add together when reused
- (D) about 37.0 °C, because the thread always falls to normal body temperature between patients

10. Why is a clinical thermometer's scale spread over only 35–42 °C while a laboratory thermometer's scale is spread over -10 °C to 110 °C, even though both measure in Celsius?

- (A) the clinical thermometer's liquid cannot expand much, so its scale must be short
- (B) a short scale only makes the clinical thermometer cheaper, which is the sole reason
- (C) human body temperature only varies over a small band, so squeezing the scale into 35–42 °C lets the small divisions be fine enough to read a fever to a tenth of a degree
- (D) laboratory thermometers measure heat while clinical thermometers measure temperature, needing different scales

11. A student wants to watch a beaker of water rise from 15 °C all the way to boiling, reading it continuously while it heats. Which thermometer suits the task, and why is the other wrong?

- (A) a clinical thermometer, because its kink lets you read it without holding it still
- (B) a clinical thermometer, because it reads in Celsius like all thermometers
- (C) a laboratory thermometer, because its range covers the whole rise to boiling and its kink-free thread can move up and down freely while you keep reading it
- (D) either thermometer, since both read up to 110 °C

12. A laboratory thermometer in boiling water reads near 100 °C. A student tips it nearly horizontal to read it, then lifts it out into cool air. Compared with a clinical thermometer treated the same way, what happens to each reading?

- (A) both readings stay locked, because every thermometer traps its thread once it stops rising
- (B) only the laboratory thermometer holds its reading, because having no kink lets it freeze the thread
- (C) both readings climb higher after removal, because the glass stays hot for a while
- (D) the laboratory thermometer's thread falls as the liquid cools in the air, so its reading drops; a clinical thermometer's kink would instead hold its highest reading

13. Two thermometers cover 0 °C to 100 °C over the same length of glass. Thermometer X has 200 small divisions across that span; thermometer Y has 40. Which can be read more finely, and what is each smallest division worth?

- (A) thermometer Y reads more finely, because fewer divisions each cover a smaller temperature
- (B) both read equally finely, because they cover the same 0–100 °C range
- (C) thermometer X reads more finely; each of its divisions is 0.5 °C, while each of Y's is 2.5 °C
- (D) thermometer X's divisions are 2 °C and Y's are 0.5 °C

14. On a laboratory thermometer the numbered marks are 4 degrees apart, with 8 equal small divisions between consecutive numbers. The thread rests on the 5th small division above the 20 mark. What is the temperature?

- (A) 25 °C, because 5 divisions above the 20 mark simply means 25 °C
- (B) 20.5 °C, because 5 divisions above a mark always means five tenths

- (C) 22.5 °C, because each small division is 0.5 °C, so 5 divisions above 20 is $20 + 2.5$
(D) 24 °C, because 8 divisions of 0.5 °C take you straight to the next numbered mark

15. A clinical thermometer is spread from 35 °C to 42 °C, with each whole degree split into 5 equal small divisions. A patient's thread stops on the 4th small division above the 38 mark. The reading is:

- (A) 38.4 °C, because 4 divisions above 38 means four tenths of a degree
(B) 38.8 °C, because each division is 0.2 °C, so 4 divisions above 38 is $38 + 0.8$
(C) 42 °C, because 4 divisions takes you to the top of the scale
(D) 38.04 °C, because 4 divisions above 38 means four hundredths

16. A laboratory thermometer's thread climbs steadily while water heats. At 4:00 it reads the 1st division above 30; at 4:02 the 5th division above 30; at 4:04 the 9th division above 30. Each division is 1 °C, and there are 10 divisions between the 30 and 40 marks. If the steady rise continues, what does it read at 4:06?

- (A) about 40 °C, because the thread cannot rise past the 10th division above 30
(B) about 78 °C, because the temperature doubles every two minutes
(C) about 43 °C, because it rises about 2 °C every minute, reaching 31, 35, 39, then 43 by 4:06
(D) exactly 39 °C, because the thread always stops at the last division it reached

17. Two students read the same laboratory thermometer where the numbered marks are 2 degrees apart with 4 equal divisions between them. The thread sits on the 3rd division above the 26 mark. One writes 26.3 °C and the other writes 27.5 °C. Who is right and why?

- (A) 26.3 °C is right, because 3 divisions above a mark always means three tenths
(B) 27.5 °C is right — each division is 0.5 °C, so 3 divisions above 26 is $26 + 1.5$
(C) both are right, since 26.3 °C and 27.5 °C are two ways of writing the same value
(D) neither is right; 3 divisions above 26 is 29 °C

18. A laboratory thermometer is numbered every 10 degrees with 10 equal divisions between numbers. The thread sits 3 divisions below the 0 mark, in the negative region. What temperature does it read?

- (A) -3 °C, because each division is 1 °C and being 3 divisions below 0 means 3 degrees below zero
(B) 3 °C, because 3 divisions above the bottom of the scale means positive 3
(C) -30 °C, because 3 divisions of 10 degrees each take you to minus thirty
(D) -13 °C, because below 0 you count down from the -10 mark

19. A laboratory thermometer numbered every 5 degrees with 5 divisions between numbers reads the 4th division above 10 at 9:00 and the 1st division above 20 at 9:05. By how much did the temperature change, and in which direction?

- (A) it fell by 3 °C, because the second reading shows fewer divisions (1) than the first (4)
(B) it rose by 7 °C, from 14 °C to 21 °C, because each division is 1 °C

(C) it rose by 10 °C, the gap between the 10 and 20 marks

(D) it rose by 3 °C, from 14 °C to 17 °C

20. A copper rod and a wooden rod of identical shape are each heated at one end with the same flame. A wax bead on the far end of the copper rod melts in 20 seconds. The wood bead does not melt at all even after 10 minutes. The clearest conclusion about how heat moves through each rod is:

(A) heat radiates across the rods, and copper happens to radiate faster than wood

(B) hot metal rises inside the copper rod and carries heat by convection to the far end

(C) heat is conducted along both rods, but copper conducts it to the far end much faster, while wood is such a poor conductor that very little heat ever reaches its far bead

(D) the copper bead makes its own heat once the rod warms, which is why it melts

21. On a cold morning a steel spanner and a wooden ruler have both been on the same shelf all night and are at the same temperature, 8 °C. You touch each with a 32 °C hand. From which does your hand lose heat faster, and which feels colder?

(A) the wood feels colder, because wood is at a lower temperature than steel after a cold night

(B) both feel equally cold, because both are at the same 8 °C

(C) the steel feels warmer, because metal stores heat and gives some back to the hand

(D) your hand loses heat faster to the steel spanner, so the steel feels colder, even though both are at the same 8 °C

22. Why does a copper block conduct heat far faster than a same-sized block of still air, explained in terms of particles?

(A) air has no particles, so there is nothing in it to conduct heat at all

(B) copper is colder than air, and colder materials always conduct heat better

(C) air's particles move too slowly to carry any energy, so conduction stops completely

(D) copper's particles are packed closely and pass vibrations to their neighbours readily, while air's particles are far apart and rarely collide, so they hand on energy very slowly

23. A cook needs the base of a pan to heat the food fast from the burner but wants the handle to stay cool enough to grip. Which material pairing achieves BOTH goals, and why?

(A) a metal base, a good conductor that lets heat pass into the food quickly, with a wooden or plastic handle, a poor conductor that blocks heat from reaching the grip

(B) a wooden base and a metal handle, so heat is kept out of the food and led safely along the handle

(C) a metal base and a metal handle, since one material is simplest and metal handles stay coolest

(D) a plastic base and a plastic handle, so the whole pan resists heat and never gets dangerously hot

24. A barefoot child walks across a tiled floor and a wooden floor in the same room; both floors are at the same 24 °C. The tiles feel cold underfoot while the wood feels neutral. The best explanation is:

- (A) the tile is genuinely several degrees colder than the wood after sharing the room
- (B) the wood produces a little heat that warms the foot, making it feel neutral
- (C) the foot judges only tile correctly, so the wood comparison is meaningless
- (D) the tile conducts heat away from the foot faster than the wood, so the foot cools faster on the tile and it feels colder, though both are at 24 °C

25. Which single list contains only good conductors of heat, with no insulator hiding among them?

- (A) copper, wood, aluminium
- (B) iron, air, silver
- (C) aluminium, plastic, copper
- (D) copper, aluminium, iron

26. Water in a kettle is heated only at the centre of its base. Just over the flame, the water:

- (A) becomes denser and sinks, while cool water rises over it, because hot water is always the heaviest
- (B) expands, becomes less dense, and rises, while cooler denser water from the sides sinks to replace it, setting up a circulating current
- (C) stays still and conducts its heat straight up through the still layers above without moving
- (D) turns instantly to steam at the bottom and bubbles straight up without any circulation

27. A beaker is heated at one corner of its base rather than the centre. Coloured dye dropped at that corner rises there, drifts across the top, and sinks down the far side. What does this reveal about where convection currents form?

- (A) the current rises directly over wherever the water is heated and sinks where it is cooler, so an off-centre heat source still drives a loop rooted at that corner
- (B) the current always rises in the exact centre of the beaker, no matter where the heat is applied
- (C) there is no current; the dye only spreads by conduction through the still water
- (D) the dye sinks first at the heated corner because heating makes the water there heavier

28. A room is warmed by a heater placed near the floor. If the same heater were moved up near the ceiling instead, what would most likely happen to how well the room heats?

- (A) the room would heat faster, because heat from up high falls down to fill the room
- (B) the warm air from a ceiling-level heater would mostly stay up near the ceiling, so the lower part of the room would be poorly heated — which is why heaters are placed low

(C) nothing would change, because air carries heat equally well from any height by conduction

(D) the floor would become the hottest part, because warm air always sinks to the floor

29. A student claims, 'If I heat a solid copper bar in the middle, the warm part will rise and the cool parts will sink, setting up a convection current inside the bar just like in a kettle.' Why is this impossible?

(A) it is possible, but only after the copper bar is painted black to absorb the heat

(B) it is possible, because any material convects once it is hot enough

(C) convection needs a fluid whose heated parts can physically move and circulate; in a solid bar the particles are fixed in place and cannot flow, so heat moves only by conduction

(D) it is impossible because a solid bar cannot be heated in the middle at all

30. Two clear tubes are set up. In tube P, hot coloured water is gently floated on top of cold clear water; in tube Q, hot coloured water is gently placed at the bottom under cold clear water. In which tube does a strong convection current develop, and why?

(A) in tube P, because hot water on top always sinks and starts a strong current

(B) in both tubes equally, because any temperature difference always drives convection

(C) in tube Q, because the hot water at the bottom is lighter and rises while the cold water sinks, driving circulation; in tube P the hot water is already on top, the stable arrangement, so little current forms

(D) in neither tube, because water cannot convect once it is coloured

31. Sitting beside a campfire on a still night, you feel warmth on the side of your face turned toward the flames, even though the hot air streams straight up. If a sheet of clear glass is held between your face and the fire, you still feel some warmth come through. This shows the warmth reaches you mainly by:

(A) convection, because hot air must be curling around the glass to reach your face

(B) conduction, because the glass touches the air which touches your face in a chain

(C) the fire pushing the cold night air toward you through the glass

(D) radiation, which travels in straight lines, needs no moving air, and can pass through transparent glass to reach your skin

32. Two identical black metal plates sit in the same cool dark room, one at 500 °C and one at 150 °C. Comparing how much heat each gives off as radiation every second:

(A) the 150 °C plate radiates more, because cooler objects shed heat faster to catch up

(B) both radiate the same amount, because they are identical plates of the same colour and size

(C) the 500 °C plate radiates much more heat per second, because the hotter an object is, the more radiation it emits

(D) neither radiates, because radiation only happens in direct sunlight

33. A space probe drifting far from any star, surrounded by the vacuum of space, slowly cools toward the cold of space. With no air or matter around it, by which mode does it lose heat, and why are the others impossible?

- (A) by convection, since hot gases stream off the probe into the surrounding space
- (B) by conduction, since the vacuum carries heat directly away from the surface
- (C) it cannot lose any heat at all, because all three modes need matter and a vacuum has none
- (D) it loses heat by radiation, because radiation needs no medium; conduction needs touching particles and convection needs a flowing fluid, neither of which exists in a vacuum

34. Two identical kettles of equally hot water cool in the same cool room: kettle M has a dull, dark, matt surface; kettle N has a bright, shiny, polished surface. After twenty minutes, which kettle's water is cooler, and why?

- (A) kettle M (dull dark surface) is cooler, because dull dark surfaces are better emitters of radiation and lose heat faster
- (B) kettle N (shiny surface) is cooler, because shiny surfaces radiate heat away fastest
- (C) both are equally cool, because they hold the same water at the same start temperature
- (D) neither cools by radiation indoors, because radiation needs direct sunlight

35. A shiny silver teapot keeps tea hot longer than an identical dull black teapot, yet on a sunny day a dull black object warms up faster in sunlight than a shiny one. Which single principle explains BOTH observations?

- (A) dull dark surfaces are both better emitters AND better absorbers of radiation, while shiny surfaces are poorer at both — so black loses heat faster but also gains heat faster
- (B) shiny surfaces both absorb and emit radiation better, which is why they keep tea hot and warm in sunlight
- (C) dark surfaces absorb well but cannot emit, while shiny surfaces emit well but cannot absorb
- (D) colour affects only absorption and never emission, so the teapot result must be due to something else

36. A student lists three claims about a perfect vacuum: (1) conduction can cross it, (2) convection can cross it, (3) radiation can cross it. Exactly how many are correct, and which?

- (A) exactly two, because conduction and radiation both cross a vacuum but convection cannot
- (B) exactly one — only claim (3), since radiation needs no medium, while conduction needs touching particles and convection needs a flowing fluid, both absent in a vacuum
- (C) all three, because heat always finds some way across any gap
- (D) none, because every mode of heat transfer needs matter to travel through

37. A vacuum flask keeps soup hot using a near-vacuum gap between two walls AND shiny silvered inner surfaces. Match each feature to the mode of heat loss it chiefly fights:

- (A) the vacuum blocks radiation, while the shiny walls block conduction and convection
- (B) the vacuum gap blocks conduction and convection because both need matter to cross, while the shiny silvered walls cut radiation because shiny surfaces are poor emitters
- (C) the vacuum blocks all three modes by itself, and the shiny walls only make it look nice
- (D) the shiny walls block all three modes, and the vacuum gap only adds strength

38. A foam cup, a thin metal cup, and a ceramic mug of the same shape each hold the same hot drink in the same room. Ranking how well they keep the drink hot, from best to worst, the most reasonable order is:

- (A) thin metal best, foam worst, because metal seals the heat in by surrounding the drink
- (B) all three keep it equally hot, since they hold the same drink in the same room
- (C) foam best, ceramic in the middle, thin metal worst — foam's trapped air is the poorest conductor, while metal conducts heat away fastest
- (D) ceramic best, foam worst, because solid walls block heat better than air pockets

39. Why is air, all around us, often the best insulator in clothing yet a poor insulator on a windy day?

- (A) still trapped air conducts heat very poorly and so insulates well, but moving air carries heat away by convection, so wind makes air a poor insulator
- (B) air is always a perfect insulator whether still or moving, because it has so few particles
- (C) moving air conducts heat better than still air, which is why wind cools you
- (D) air destroys heat when still but creates heat when moving

40. A shiny silver-coloured emergency blanket is wrapped around a cold hiker. Unlike a thick wool blanket which works by trapping air, the shiny blanket keeps the hiker warm chiefly by:

- (A) generating warmth from the silver coating once it touches the skin
- (B) trapping a very thick layer of air inside the thin silver sheet
- (C) conducting the cold surroundings away from the hiker quickly
- (D) reflecting the hiker's own radiated body heat back toward them instead of letting it escape

41. A market vendor stands in bright winter sunshine. Between an otherwise identical white coat and black coat, which feels warmer in the sun, and what is the night-time trade-off?

- (A) the white coat feels warmer in the sun, because pale colours absorb more sunlight
- (B) the black coat feels warmer in the daytime sun because it absorbs more solar radiation; but at night, away from the Sun, the black coat would also emit body heat away faster
- (C) the black coat is warmer at every time of day, because black always makes more heat
- (D) colour makes no difference in the sun, since both coats are the same thickness

42. Two sealed boxes with thermometers sit in identical bright sunlight through glass lids: one lined dull black inside, one lined shiny white. After an hour the black-lined box reads much higher. Both are then moved into a dark cool room. Which box's thermometer falls fastest, and why?

- (A) the shiny white box cools fastest, because shiny surfaces lose heat by radiation fastest
- (B) both cool at the same rate, because they are the same size and shape
- (C) the black-lined box cools fastest, because a dull dark surface is also a better emitter and radiates its heat away faster than the shiny one
- (D) neither cools, because once heated a sealed box keeps its heat forever

43. A solar water heater on a roof uses a metal absorber plate behind glass to warm water in pipes. Why is the absorber plate painted dull black rather than shiny white?

- (A) a dull black surface absorbs the most of the Sun's radiation, heating the plate and the water in the pipes most effectively
- (B) a dull black surface reflects the most sunlight, sending the heat into the pipes by reflection
- (C) a shiny white surface would absorb more, but black is chosen only because it is cheaper
- (D) the colour of the plate does not matter, since the glass alone does all the heating

44. On a sunny afternoon at the coast a windsock points steadily inland, from the sea toward the land. Combining the daytime pattern with its cause, this is a:

- (A) land breeze: the sea heats faster than the land by day, pushing air toward the shore
- (B) sea breeze caused by the Sun directly dragging air landward across the water
- (C) sea breeze: the land heats faster than the sea by day, so air rises over the warm land and cooler air flows in from over the sea
- (D) land breeze caused by the land staying cooler than the sea all afternoon

45. Well after sunset, a sailor near the coast finds a steady wind blowing from the shore out to the open sea. Identify the breeze and the temperature situation driving it:

- (A) a sea breeze: the sea has cooled faster than the land, so air flows toward the open sea
- (B) a land breeze: the land stays warmer than the sea at night, pushing air seaward

(C) a land breeze: the land has cooled faster than the sea, so the sea is now warmer; air rises over the warm sea and cooler air flows out from the land

(D) a sea breeze caused by the night air automatically reversing the daytime wind

46. A coastal observer logs wind blowing inland from about 10 a.m. to 6 p.m. and out to sea from about 10 p.m. to 6 a.m., with quiet, shifting winds near dawn and dusk. The calm periods near dawn and dusk happen because:

(A) near dawn and dusk the sea and land are at their greatest temperature difference

(B) near dawn and dusk the land and sea are at nearly the same temperature, so there is little difference to drive air in either direction

(C) the Sun is directly overhead at dawn and dusk, which cancels all breezes

(D) the wind always stops for exactly two hours twice a day by a fixed natural rule

47. A geography teacher says a sea breeze and a land breeze are 'the same convection process running in opposite directions.' Why is the teacher correct?

(A) the sea breeze is convection but the land breeze is conduction through the cooler ground at night

(B) the sea breeze is driven by the Sun and the land breeze by the Moon, so they are different forces

(C) they are different because the sea breeze is radiation and the land breeze is convection

(D) both are convection: air rises over whichever surface is warmer and cooler air flows in from the cooler surface; by day the land is warmer, by night the sea is warmer, so the same process simply reverses

48. On an unusual, perfectly even-temperature coast where the land and sea always stay at exactly the same temperature day and night, would a sea breeze or a land breeze blow? Why?

(A) neither — with no temperature difference, no surface is warmer to make air rise, so there is nothing to drive a breeze in either direction

(B) a sea breeze, because the sea is always cooler than the land regardless of temperature

(C) a land breeze, because nights always send air out to sea no matter the temperatures

(D) both at once, because a coast always produces some breeze

49. Trace this chain on a sunny coastal afternoon: (1) the dark land absorbs sunlight and warms faster than the sea; (2) the air just above the warm land is heated and rises; (3) cooler sea air flows in to replace it. Name the heat-transfer mode in step 1 and in steps 2–3, in that order:

(A) step 1 is conduction (the Sun conducts heat to the land); steps 2–3 are radiation (the warm air beams to the sea)

(B) step 1 is radiation (the Sun's heat crossing space and absorbed by the land); steps 2–3 are convection (warm air rising and cooler air flowing in)

(C) step 1 is convection (hot solar gas reaches the land); steps 2–3 are conduction (the air conducts heat sideways)

(D) all three steps are conduction, since the ground, air, and sea are all in contact

50. A student lists how heat reaches them in three situations and the mode involved:

'Hands warmed by gripping a hot mug — conduction. Warmth felt across a room from a glowing heater I am not touching — radiation. Warm air spreading from a floor heater to fill the room — convection.' How many of these three pairings are correct?

(A) only two are correct

(B) only one is correct

(C) none is correct

(D) all three are correct

51. A thermos of hot cocoa with vacuum walls and a shiny lining is left in a cold room for hours and eventually goes lukewarm. Which statement best explains why a thermos only SLOWS heat loss rather than stopping it forever?

(A) the vacuum and shiny walls greatly slow conduction, convection, and radiation, but the cap, seams, and a little radiation through the walls still let heat leak slowly, so it cannot be perfect

(B) a thermos makes a little heat of its own, which simply runs out after a few hours

(C) a vacuum actually speeds up heat loss, so the cocoa cools as fast as in any open cup

(D) cold from the cold room flows into the cocoa through the vacuum, which the thermos cannot block

52. On a freezing night a thin steel sheet and a thick foam board are left outdoors together and both reach the air temperature of 1 °C. A barefoot person steps briefly on each. Which feels colder underfoot, and would a thermometer pressed on each read differently?

(A) the foam feels colder, and a thermometer would read a lower temperature on the foam

(B) the steel feels much colder underfoot because it conducts heat from the foot quickly, yet a thermometer would read about 1 °C on both since they are at the same temperature

(C) both feel equally cold, and a thermometer reads 1 °C on both, since feel and temperature always match

(D) the steel feels colder, and a thermometer would also read a lower temperature on the steel, proving it is genuinely colder

53. 300 g of water at 70 °C is mixed with 100 g of water at 30 °C in an insulated jar. Using the idea that the larger share dominates and equal amounts would meet midway, the final temperature is closest to:

(A) 50 °C, the plain midpoint of 30 and 70, because the temperatures simply average

(B) 40 °C, nearer the cold value, because cold water always pulls the mix down hardest

(C) 60 °C, nearer the hot value, because there is three times as much hot water as cold, so the mix settles much closer to 70 °C than to the 50 °C midpoint

(D) 100 °C, because the two temperatures (70 and 30) combine to make the total

54. A clinical thermometer reads only about 35–42 °C. A doctor wants to record the temperature of a furnace-heated metal sample at about 150 °C. Why is the clinical thermometer the wrong tool, even setting the kink aside?

- (A) 150 °C is far above the clinical thermometer's top mark of about 42 °C, so its scale cannot reach that temperature at all
- (B) its liquid would freeze solid at 150 °C, giving no reading
- (C) its kink would hold the reading too low to be useful at high temperatures
- (D) it reads in Celsius, but furnace samples must be measured in different units

55. A laboratory thermometer reads 22 °C in a room. A student grips its bulb in a warm hand until the thread rises to 34 °C, then lets go and watches. What does the thread do, and what does it show about how the thermometer works?

- (A) the thread stays at 34 °C, because a thermometer always keeps its highest reading
- (B) the thread falls back toward 22 °C, showing the liquid expands when warmed by the hand and contracts again as it cools, with no kink to hold it
- (C) the thread keeps rising past 34 °C, because the hand keeps adding heat after letting go
- (D) the thread drops to 0 °C, because a thermometer with no hand on it always reads zero

56. On a bright sunny day, smoke from a small fire over a dark patch of bare ground climbs almost straight up, while smoke over a nearby shaded grassy patch barely rises. The difference is best explained by:

- (A) smoke is naturally lighter over bare ground than over grass, regardless of any heating
- (B) conduction through the ground physically pushes the smoke upward over the dark patch
- (C) radiation from the bare ground grabs the smoke particles and lifts them straight up
- (D) the dark bare ground gets hotter, warming the air above it more, so that air rises faster as a stronger convection current, lifting the smoke higher

57. A whitewashed house and an identical dark-painted house, same thick walls and windows, stand side by side in a hot desert town. By mid-afternoon, which is cooler inside, and why?

- (A) the dark house is cooler inside, because dark colours keep buildings cool by absorbing heat away from the rooms
- (B) the whitewashed house is cooler inside, because its light surface reflects most of the Sun's radiation, so the walls absorb less heat
- (C) both are equally cool inside, because the walls are the same thickness
- (D) the whitewashed house is hotter inside, because white absorbs the most sunlight

58. Order these four cups of identical hot tea in the same room, from the one that stays hot LONGEST to the one that cools soonest: (i) a foam cup with a lid, (ii) a plain foam cup,

(iii) a thin metal cup wrapped in wool, (iv) an uncovered thin metal cup. The correct order is:

- (A) uncovered metal cup longest, then plain foam cup, then wool-wrapped metal cup, then foam-with-lid soonest
- (B) all four cool at the same rate, since they hold the same tea in the same room
- (C) plain foam cup longest, then foam-with-lid, then uncovered metal, then wool-wrapped metal soonest
- (D) foam cup with a lid longest, then wool-wrapped metal cup, then plain foam cup, then uncovered metal cup soonest

59. A laboratory thermometer numbered every 2 degrees with 4 divisions between numbers reads the 1st division above the 18 mark in the morning and the 3rd division above the 22 mark in the afternoon. By how much did the temperature change?

- (A) it rose by 2 °C, because the afternoon division count (3) is two more than the morning's (1)
- (B) it rose by 4 °C, the gap between the 18 and 22 marks
- (C) it rose by 5 °C, from 18.1 °C to 23.1 °C, because each division is a tenth
- (D) it rose by 5 °C, from 18.5 °C to 23.5 °C, because each division is 0.5 °C

60. A student claims, 'The Sun warms the Earth by conduction through the layers of air between them.' Why is this wrong, and what is the true mode?

- (A) between the Sun and Earth lies the vacuum of space with no matter to conduct through, so the Sun's heat reaches us by radiation, which needs no medium
- (B) it is right, because the air between the Sun and Earth conducts the heat down to us
- (C) it is wrong; the true mode is convection, with hot gases flowing from the Sun to the Earth
- (D) it is wrong; the true mode is conduction, but only through the thin upper air, not radiation

Solutions — Science (Class 7), Chapter 4: Heat — Paper 5

Paper 5 — (progressively harder)

Marking: +4 correct, −1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	11	C	21	D	31	D	41	B	51	A
2	C	12	D	22	D	32	C	42	C	52	B
3	A	13	C	23	A	33	D	43	A	53	C
4	D	14	C	24	D	34	A	44	C	54	A
5	B	15	B	25	D	35	A	45	C	55	B
6	B	16	C	26	B	36	B	46	B	56	D
7	A	17	B	27	A	37	B	47	D	57	B
8	A	18	A	28	B	38	C	48	A	58	D
9	A	19	B	29	C	39	A	49	B	59	D
10	C	20	C	30	C	40	D	50	D	60	A

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (D) The two samples reach one common temperature. Because there is three times as much cold water as hot, the balance point is pulled strongly toward the cold side, not to the plain midpoint: the 80 °C gap is split so that the cold side, being three parts to one, takes only a quarter of the rise. The hot 100 g falls about 60 degrees (from 90 to 30) while the cold 300 g rises about 20 degrees (from 10 to 30) — equal total heat moved, landing near 30 °C. The plain-midpoint answer ignores the unequal amounts. The 'much nearer the hot value' answer wrongly assumes the hotter water wins regardless of quantity; here the cold share is bigger. Temperatures of mixed water never add together, so 100 °C is impossible — that would be hotter than either sample.

2. (C) Equal masses do not mean equal share of the gap here, because water takes much more heat to warm one degree than the metal gives up cooling one degree. So the metal cools through a large range while the water warms only a little; the meeting point lands well below the 60 °C midpoint and near the water's 20 °C. The equal-mass midpoint answer ignores that water and metal respond very differently to the same heat. The 'near

100 °C' answer wrongly lets the hotter object dominate; in fact the water barely shifts. And water certainly does change temperature when heat enters it — it simply rises slowly, so it does not stay exactly at 20 °C.

3. (A) Temperature tells you how hot each liquid ended up; it does not by itself tell you how much heat energy each one swallowed. Equal final temperature with equal volumes does not force equal heat absorbed, since different liquids warm by different amounts for the same heat. The 'always equal heat' answer confuses equal temperature with equal heat. Oil is not 'always hotter' — both ended at the same temperature by the question's wording. And heat and temperature are related, not unrelated: heat is the energy that flowed in, temperature is the resulting hotness, so 'no heat absorbed' is plainly wrong since both warmed up.

4. (D) Heat flows only from a hotter place to a cooler one; with everything at 25 °C there is no temperature difference, so there is no net flow at all. The tank does hold more total heat than the coin, but total heat content does not drive flow — only a temperature gap does, and there is none. The idea that small objects always lose heat to large ones, or that rooms always warm metal coins, both assume a temperature difference that this question removes. Same temperature everywhere means equilibrium and no net heat transfer.

5. (B) The mug started much colder than the tea, so heat flowed from the hotter tea into the cooler mug until they approached the same temperature; warming a cold mug takes a fair amount of heat, which is why the tea drops quickly at first. There is no such thing as 'cold flowing in' — only heat flows, and it flows out of the tea, not cold into it. The warm room cannot cool the tea by adding warmth; if anything the room is cooler than 70 °C tea, so the room also receives heat. And tea does not 'make heat' — it only holds the heat it had and loses it; its temperature falls because heat leaves it, chiefly into the cold mug here.

6. (B) For equal amounts of the same liquid, the heat taken in depends on the size of the temperature rise, not the final value. P climbs 15 degrees (10 to 25) while Q climbs only 10 degrees (50 to 60), so P needed more heat. Ending at a higher temperature (Q at 60 °C) does not mean more heat was added — Q started high and rose little. 'Same amount of water' alone is not enough; the rises differ, so the heat differs. And heat certainly entered both samples since both warmed; different starting temperatures do not block heat from entering.

7. (A) Heat is conserved: the heat the ball loses equals the heat the water gains. But the two change by different numbers of degrees because water resists temperature change far more strongly than the metal does — so the same heat moves the ball's temperature far more than the water's. Heat does not 'disappear', so the loss does not exceed the gain. The two do not change by equal degrees; only the heat amounts are equal, not the temperature swings. And metal does not 'lose all its temperature instantly' — it cools gradually until it matches the water.

8. (A) Only the laboratory thermometer reaches -6°C , since the clinical scale starts at 35°C — so the brine must use the lab thermometer. The 38.4°C bath reading must be carried across the room, which needs the kink that holds the thread: that is the clinical thermometer's job. Using the clinical thermometer for everything fails on the brine, which is far below its range. Using the laboratory thermometer for everything fails the 'carry the reading away' need, because its thread falls back the moment it leaves the warm water. The last option reverses both tools: a clinical thermometer cannot reach -6°C , and a lab thermometer cannot hold a reading while carried.

9. (A) The kink in a clinical thermometer keeps the mercury thread from slipping back, so it stays at the highest value reached until shaken down. Since 40.1°C is higher than the second child's 38.5°C , the cooler body cannot push the thread up, and it cannot fall on its own — so she still reads about 40.1°C . The thermometer does not 'reset' itself; that is exactly why it must be shaken. Readings never add together. And the thread does not drift to 37°C on its own — without a shake it stays where the kink left it.

10. (C) Because the body only ranges over a few degrees, a clinical thermometer can devote its whole scale to $35\text{--}42^{\circ}\text{C}$, spreading those degrees out so each tiny division is small and a fever can be read very finely. The liquid's ability to expand is not the limit — the same kind of liquid is used in laboratory thermometers over a far wider range. Cost is not the reason; the design is about precision in the body range. And both thermometers measure temperature, not 'heat versus temperature' — that distinction is not what sets their ranges.

11. (C) Tracking water to boiling needs a scale reaching about 100°C and a thread that follows the temperature freely as it rises — exactly the laboratory thermometer. A clinical thermometer's range stops near 42°C , so it cannot reach boiling, and its kink would lock the highest reading instead of following the changing temperature. Reading in Celsius does not help if the range is too small. And the two thermometers are not interchangeable: only the laboratory one reaches 110°C , so 'either' is wrong.

12. (D) A laboratory thermometer has no kink, so once it leaves the hot water and cools in air, its liquid contracts and the thread falls — the reading changes. A clinical thermometer's kink would hold the highest value, so it would keep the reading. So the two behave oppositely, not identically — 'both locked' is wrong. Having no kink does not freeze a thread; it does the opposite, letting it move freely. And a cooling thermometer's thread falls, it does not climb, so 'both climb after removal' is wrong.

13. (C) Across a 100-degree span, X's 200 divisions each cover $100 \div 200 = 0.5^{\circ}\text{C}$, while Y's 40 divisions each cover $100 \div 40 = 2.5^{\circ}\text{C}$. The smaller a division, the finer the reading, so X (0.5°C steps) reads more finely than Y (2.5°C steps). Fewer divisions over the same range means each one covers MORE temperature, not less, so Y is coarser, not finer. Sharing the same range does not make them equally fine — the division count decides that. The last option swaps the two values, which is arithmetically wrong.

14. (C) The gap between numbers is 4 degrees split into 8 divisions, so each division is $4 \div 8 = 0.5$ °C. Five divisions above 20 is $5 \times 0.5 = 2.5$, giving $20 + 2.5 = 22.5$ °C. Counting '5 divisions = 5 degrees' wrongly treats each division as a whole degree. Treating divisions as automatic tenths gives 20.5 °C, but here each division is half a degree, not a tenth. The 24 °C value is the next numbered mark (8 divisions up), but the thread is only 5 divisions up, not 8.

15. (B) Each whole degree is split into 5 divisions, so each division is $1 \div 5 = 0.2$ °C. Four divisions above 38 is $4 \times 0.2 = 0.8$, giving $38 + 0.8 = 38.8$ °C. Reading the divisions as tenths gives 38.4 °C, but here they are fifths (0.2 °C each), not tenths. Four divisions is not the top of the scale — 42 °C is far higher. And 38.04 °C wrongly treats the divisions as hundredths.

16. (C) The readings are 31 °C (4:00), 35 °C (4:02), 39 °C (4:04) — a rise of 4 °C every two minutes, i.e. about 2 °C per minute. Continuing at that rate, by 4:06 (two more minutes) it adds 4 °C to 39, reaching about 43 °C. The scale does not stop at 40; the numbered marks continue above 40, so the thread can pass the 10th division. Heating at a steady rate adds equal amounts, it does not double, so 78 °C is wrong. And a rising thread does not freeze at 39 °C; it keeps climbing as the water keeps heating.

17. (B) Two degrees split into 4 divisions makes each division $2 \div 4 = 0.5$ °C. Three divisions above 26 is $3 \times 0.5 = 1.5$, so $26 + 1.5 = 27.5$ °C. Reading divisions as tenths gives 26.3 °C, but here each is half a degree, not a tenth — that student misread the scale. The two values 26.3 and 27.5 are clearly different numbers, so 'both right' is impossible. And 29 °C would require 3 whole degrees, not 3 half-degree divisions, so 'neither' is wrong too.

18. (A) Ten degrees split into 10 divisions makes each division 1 °C. Three divisions below the 0 mark is 3 degrees below zero, i.e. -3 °C. The reading is below zero, so it must be negative, not $+3$ °C. Each division is 1 °C, not 10 °C, so -30 °C overcounts each step tenfold. And the thread is measured from the 0 mark (3 below it), not from the -10 mark, so -13 °C uses the wrong reference point.

19. (B) Five degrees split into 5 divisions makes each division 1 °C. The first reading is $10 + 4 = 14$ °C; the second is $20 + 1 = 21$ °C. The change is $21 - 14 = 7$ °C, a rise. Comparing only the division counts (4 versus 1) misses that the second reading sits above the higher 20 mark, so the temperature rose, not fell. The 10-degree value is just the gap between numbered marks, not the actual change. And 17 °C would be the 2nd division above 15, which is not where either reading sits.

20. (C) In a solid rod heat travels by conduction — particle to particle along the metal — and copper passes it on far more readily than wood, so the copper bead melts quickly while the wood's bead is barely warmed. The rods are solids, so radiation is not the main mechanism inside them. Convection needs a fluid that can flow; the particles of a solid rod are fixed, so hot metal cannot 'rise' through it. And the wax bead does not generate heat; it

only melts when conducted heat arrives, which happens fast in copper and hardly at all in wood.

21. (D) Steel conducts heat away from your warm hand much faster than wood, so your skin cools faster touching the steel and it feels colder — even though a thermometer would read 8 °C on both. The two objects are at the same temperature by the question's wording, so the wood is not actually colder. 'Feel' depends on how fast heat leaves your hand, not on temperature alone, so they do not feel equally cold despite matching temperatures. And the steel does not give heat back to a 32 °C hand — your hand is warmer, so heat flows from hand into steel, making it feel cold.

22. (D) In copper the particles are close together and vibrate against their neighbours, passing heat quickly from particle to particle. In air the particles are spread far apart and collide only rarely, so energy is handed on very slowly — air conducts poorly. Air is not empty; it does contain particles, just widely spaced. A material's conducting ability is not decided by being colder — copper is the better conductor regardless of temperature. And air's particles do move; conduction in air is slow, not totally stopped — air still conducts a little, just very poorly.

23. (A) A metal base conducts the burner's heat readily into the food, while a wood or plastic handle conducts poorly, blocking heat from the grip so it stays cool — both goals met. A wooden base would block heat from reaching the food, defeating the cooking, and a metal handle would carry heat to the hand and burn it. A metal handle does not 'stay coolest'; metals conduct heat to the hand quickly. And a plastic base would resist passing heat into the food, so the food would not cook well.

24. (D) Tile is a better conductor than wood, so it draws heat from the warm foot more quickly; the faster cooling of the skin makes the tile feel cold even though a thermometer reads 24 °C on both. The two are at the same temperature by the question's wording, so the tile is not actually colder. Wood does not make its own heat — it just conducts the foot's heat away slowly, so the foot stays warmer and the wood feels neutral. And the foot judges both surfaces by the same rule — how fast heat leaves it — so the comparison is valid, not meaningless.

25. (D) Copper, aluminium, and iron are all metals and all good conductors, so this list has no insulator in it. The 'copper, wood, aluminium' list hides wood, which is a poor conductor (an insulator). The 'iron, air, silver' list hides air, which is one of the best insulators when trapped. The 'aluminium, plastic, copper' list hides plastic, a poor conductor. Only the all-metal list qualifies.

26. (B) Heated water expands and becomes less dense, so it rises; cooler, denser water from the sides sinks to take its place, and this rising-and-sinking sets up a convection current. Hot water is lighter than cold water, not 'the heaviest', so it rises rather than sinks. Water heats mainly by this moving convection, not by still conduction straight up — water

is a poor conductor. And the water does not instantly boil to steam at the base; well below boiling it simply circulates as warm and cool water trade places.

27. (A) The dye shows warm water rising directly above the heated corner and cooler water sinking elsewhere, so the convection loop forms over the actual heat source, wherever that is. The rising point follows the heat, so it is not fixed at the centre. There clearly is a current — the dye traces a loop, not still spreading — so conduction alone cannot explain it. And heating makes water lighter, not heavier, so it rises at the heated corner rather than sinking there.

28. (B) Warm air rises, so a low heater sends warmth up through the whole room as a convection current; a ceiling heater's warm air simply stays up high, leaving the lower room cool. Warm air does not 'fall down' — it rises, so a high heater does not heat faster. Air carries room heat mainly by convection currents, not by conduction, and those currents depend on the heater's height, so position matters. And warm air rises rather than sinks, so the floor does not become the hottest part under a ceiling heater.

29. (C) Convection requires a fluid — a liquid or gas — whose warmed portions can rise and circulate; in a solid copper bar the particles are locked in place and cannot flow, so heat spreads only by conduction. Painting it black affects radiation absorption, not whether a solid can convect. No material convects merely by being hot — it must be able to flow, which solids cannot. And a solid bar certainly can be heated in the middle; the point is only that the resulting heat spreads by conduction, not convection.

30. (C) Convection runs strongly when hot, lighter water sits below cold, denser water: in tube Q the hot bottom water rises and the cold top water sinks, so a vigorous current forms. In tube P the hot water is already on top — the stable, lighter-above-heavier arrangement — so there is no buoyancy push and little circulation. Hot water on top does not sink; it is lighter and stays up, so P does not drive a strong current. A temperature difference alone is not enough; the hot layer must be positioned below to rise. And colouring the water does not stop it from convecting.

31. (D) The hot air rises away from you, yet you still feel warmth — and it passes through clear glass — which is the signature of radiation: it travels straight from the fire, needs no moving air, and crosses transparent glass. If it were convection, blocking the air with glass would cut the warmth, but it does not. Conduction through a long air-to-glass-to-face chain is far too slow and weak to explain the felt warmth across a gap. And the fire does not 'push cold air' at you to warm you — that would cool you, not warm you.

32. (C) A hotter object emits more radiation each second, so the 500 °C plate radiates much more than the 150 °C one. Cooler objects do not radiate faster 'to catch up' — being cooler means emitting less. Although the plates are identical in colour and size, their temperatures differ, and temperature strongly controls how much they emit, so they do

not radiate equally. And all warm objects radiate heat, in the dark as well as in sunlight — radiation does not require sunlight to occur.

33. (D) In a vacuum there is no matter, so conduction (which needs touching particles) and convection (which needs a flowing fluid) cannot work; only radiation, which travels without any medium, can carry heat away — so the probe cools by radiation. There is no surrounding fluid in a vacuum for convection to stream into. A vacuum has no particles, so it cannot conduct. And the probe does lose heat — radiation works without matter, so it is not stuck keeping all its heat forever.

34. (A) A dull dark surface is a better emitter of radiation than a shiny one, so kettle M radiates its heat away faster and its water cools more. Shiny surfaces are poor emitters, so N loses heat more slowly and stays warmer — it is not the cooler one. Equal starting water does not mean equal cooling, because the surfaces emit at different rates. And radiation happens indoors too; all warm objects radiate heat whether or not sunlight is present.

35. (A) A dull dark surface is good at both emitting and absorbing radiation: it emits its own heat fast (so the black teapot cools quickest, and the shiny one keeps tea hot) and it absorbs sunlight fast (so the black object warms fastest in the sun). One property — that dark surfaces are strong at both emitting and absorbing — explains both results. Shiny surfaces are poor at both, the opposite of the claim that they 'absorb and emit better'. Dark surfaces can both absorb and emit; they do not 'absorb but not emit'. And colour affects emission as well as absorption, so the teapot result is fully explained, not unrelated.

36. (B) Only radiation crosses a vacuum, because it needs no medium; conduction needs particles in contact and convection needs a moving fluid, and a vacuum has neither — so just claim (3) is correct, one in total. Conduction cannot cross a vacuum, so 'two correct' is wrong. Heat does not always find a way — across a vacuum only radiation works, so 'all three' is wrong. And radiation does cross a vacuum without matter, so 'none correct' is also wrong.

37. (B) Conduction and convection both need matter, so removing the matter in the gap (the vacuum) blocks them; radiation needs no matter, so the vacuum cannot stop it — instead the shiny silvered walls, being poor emitters, cut the radiation. So vacuum handles conduction and convection, shiny walls handle radiation. The vacuum does not block radiation (radiation crosses a vacuum freely), so the reversed pairing is wrong. The vacuum cannot block radiation by itself, so the shiny walls are not merely decorative. And the shiny walls do not block conduction or convection — that is the vacuum's job — so 'shiny walls block all three' is wrong.

38. (C) Foam is full of trapped air, a very poor conductor, so it slows heat loss best; metal is a good conductor, so it carries heat away fastest and keeps the drink hot worst; ceramic falls in between. Metal does not 'seal heat in' — it conducts heat out quickly, so it is the worst, not the best. The three are made of different materials with very different

conducting abilities, so they do not keep heat equally. And solid ceramic walls do not beat foam's air pockets — trapped air is the better insulator, so foam wins, not ceramic.

39. (A) Trapped, still air conducts heat poorly, so it insulates well in clothing; but when air can move, it carries heat away by convection, so a windy day strips heat from the body and air no longer insulates. Air is not 'always a perfect insulator' — once it moves, convection defeats it. Wind cools you by carrying heat away in moving currents (convection), not because moving air suddenly conducts better. And air neither destroys nor creates heat; it only slows or speeds heat loss depending on whether it is still or moving.

40. (D) A shiny silver surface is a poor emitter and a good reflector of radiation, so the thin foil blanket bounces the hiker's own radiated body heat back instead of letting it escape — that is its main trick, different from wool's trapped air. The silver does not 'generate warmth'; no cloth makes heat. The sheet is thin and traps very little air, so it does not work the way wool does. And conducting the cold surroundings 'away' is not how it warms — the point is to keep the body's radiated heat from leaving by reflecting it back.

41. (B) Dark surfaces absorb radiation well, so the black coat soaks up more sunlight and feels warmer by day; but dark surfaces also emit well, so at night the black coat sheds the body's own heat faster — that is the trade-off. White reflects sunlight rather than absorbing it, so the white coat is the cooler one in the sun, not the warmer. Black does not 'make heat'; it only absorbs the Sun's radiation, which helps in daylight but not at night. And colour clearly matters in the sun even at equal thickness, because dark and light surfaces absorb sunlight very differently.

42. (C) A dull dark surface both absorbs well (so the black box got hotter in the sun) and emits well (so in the dark room it radiates its heat away fastest and cools quickest). Shiny surfaces are poor emitters, so the white box loses heat slowly — it does not cool fastest. Same size and shape does not make them cool equally, because their surfaces emit at very different rates. And a sealed box does not keep heat forever; both cool by radiation, just at different speeds.

43. (A) Dark surfaces absorb radiation strongly, so a dull black plate soaks up the most sunlight and passes that heat to the water in the pipes — that is why it is painted black. Black absorbs sunlight; it does not reflect the most (a white surface reflects more), so 'reflects then heats by reflection' is wrong. White absorbs less, not more, so the claim that white would absorb more is backwards. And the plate's colour matters a great deal — a white plate would reflect much of the Sun's heat away, so the glass alone is not what does the heating.

44. (C) By day the land heats faster than the sea, so warm air rises over the land and cooler sea air flows in to replace it — a sea breeze blowing from sea to land. The sea does not heat faster than the land by day; the land does, so 'land breeze, sea heats faster' is

doubly wrong. The Sun does not 'drag air' across the water; the breeze comes from the rising warm air over the land. And the land does not stay cooler than the sea in the afternoon — it is the warmer surface, which is what drives the inland sea breeze.

45. (C) At night the land loses heat faster than the sea, so the sea becomes the warmer surface; warm air rises over the sea and cooler air flows out from the land — a land breeze from land to sea. The sea does not cool faster than the land at night; the land cools faster, so 'sea cooled faster' is wrong. The land is the cooler, not warmer, surface at night, so 'land stays warmer' is wrong. And the reversal is not 'automatic' — it is caused by the land cooling faster than the sea, changing which surface is warmer.

46. (B) Breezes are driven by the temperature difference between land and sea; near dawn and dusk the two are changing over and briefly nearly equal, so there is almost no difference to push air, and the wind goes calm. The difference is smallest, not greatest, at those changeover times, so 'greatest difference' is wrong. The Sun is low, not overhead, at dawn and dusk, and its position does not 'cancel' breezes anyway. And the calm is not a fixed clock rule — it follows the land-sea temperatures, which happen to even out around those times.

47. (D) Both breezes are the same convection: warm air rises over the hotter surface and cooler air flows in from the cooler surface. Only which surface is warmer changes — land by day, sea by night — so the same process runs in reverse. The land breeze is not conduction; it is convection of moving air, just reversed. The Moon does not drive the land breeze; both breezes come from differential heating and cooling by the Sun's daily cycle. And neither breeze is radiation — moving air is convection in both cases.

48. (A) Sea and land breezes need one surface to be warmer so its air rises and cooler air flows in; with the land and sea always at the same temperature there is no warmer side, so no breeze forms in either direction. The sea is not 'always cooler' here — the two are equal by the question's wording, so no sea breeze. Nights do not 'always' send air seaward; that needs the land to cool below the sea, which cannot happen if they stay equal. And a coast does not always produce a breeze; with no temperature difference, there is no driving cause.

49. (B) The Sun's heat crosses empty space and is absorbed by the land — that is radiation (step 1). The warm air over the land then rises and cooler sea air flows in to replace it — that moving air is convection (steps 2–3). The Sun cannot conduct heat to the land across space, because conduction needs touching matter, so step 1 is not conduction. The warm air moving sideways is bulk flow, which is convection, not radiation or conduction. And not all steps are conduction; the Sun-to-land step crosses a vacuum (radiation) and the air movement is convection.

50. (D) Touching a hot mug transfers heat by direct contact, which is conduction — correct. Feeling warmth across a room from a heater you do not touch is radiation, since it

crosses the gap with no contact and no moving air needed — correct. Warm air spreading from a floor heater is moving fluid, which is convection — correct. All three match, so 'only two', 'only one', and 'none' all undercount the correct pairings.

51. (A) A thermos slows but never stops heat loss: the vacuum cuts conduction and convection and the shiny lining cuts radiation, yet the cap, the seams, and a small amount of radiation still let heat escape slowly, so the cocoa eventually cools. A thermos does not generate heat of its own, so it is not 'running out of made heat'. A vacuum slows heat loss; it does not speed it up, so the cocoa does not cool as fast as in an open cup. And 'cold' does not flow in — only heat flows, and it slowly leaks out, which is what cools the cocoa.

52. (B) Steel conducts heat away from the foot rapidly, so the skin cools fast and the steel feels much colder — but both objects are at the same 1°C , so a thermometer reads about 1°C on each. Foam conducts poorly, so it feels less cold, not more, so 'foam feels colder' is wrong. Feel and temperature do not always match — that is the whole point; the steel feels colder even though both read the same. And the thermometer would not read lower on the steel, because the steel is not genuinely colder — it only removes heat from the foot faster.

53. (C) There is three times as much hot water as cold, so the meeting point is pulled strongly toward the hot side: the cold 100 g rises about 30 degrees (from 30 to 60) while the hot 300 g falls only about 10 degrees (from 70 to 60), moving equal total heat and landing near 60°C . The plain midpoint (50°C) ignores the unequal amounts. Cold does not 'always pull hardest' — here the hot share is bigger, so the mix lands above the midpoint, not below it. And temperatures never add together, so 100°C — hotter than either sample — is impossible.

54. (A) A clinical thermometer's scale stops near 42°C , far below the 150°C sample, so it simply cannot reach that temperature — the reading would run off the top of the scale. Liquids do not 'freeze' when heated; freezing happens when cooled, not at 150°C . The kink only holds the highest value reached, but the real problem is the scale's small range, not the kink. And temperature is measured in Celsius for both — the unit is not the obstacle; the limited range is.

55. (B) A laboratory thermometer has no kink, so once the hand is removed the warmed liquid cools and contracts, and the thread falls back toward the room's 22°C — showing the liquid expands when heated and contracts when cooled. Only a clinical thermometer's kink keeps the highest reading; a laboratory one does not. Once the hand lets go it adds no more heat, so the thread cannot keep rising. And the thread settles at the room temperature it is sensing (about 22°C), not at 0°C — a thermometer reads its surroundings, not zero.

56. (D) The dark bare ground absorbs more sunlight and gets hotter, so it warms the air above it more; that warmer air rises faster as a stronger convection current and carries the smoke up. Smoke is not 'naturally lighter' over one kind of ground — what differs is how

much the ground heats the air. Conduction does not 'push' smoke upward; the rising is buoyant convection. And radiation does not grab and lift particles — it is the heating of the air, then convection, that lifts the smoke.

57. (B) A white surface reflects most of the Sun's radiation, so the whitewashed walls absorb less heat and the rooms stay cooler. Dark colours absorb more heat, warming the house, so they do not 'keep it cool' — the dark house is hotter, not cooler. Equal wall thickness does not make them equally cool, because the surface colours absorb sunlight very differently. And white reflects rather than absorbs sunlight, so the white house is cooler, not hotter.

58. (D) Foam with a lid insulates the walls and blocks the open top, so it stays hottest longest. Wool wrapped on a metal cup adds a poor-conducting trapped-air layer, beating a plain foam cup which still loses heat from its open top. The bare uncovered metal cup conducts heat away fastest, so it cools soonest. The reversed order wrongly puts the worst insulator (bare metal) first and the best (lidded foam) last. The four are different in insulation, so they do not cool at the same rate. And a plain foam cup with an open top loses more from the top than the lidded foam or the wool-wrapped metal, so it is not the best.

59. (D) Two degrees split into 4 divisions makes each division 0.5°C . Morning: $18 + (1 \times 0.5) = 18.5^{\circ}\text{C}$. Afternoon: $22 + (3 \times 0.5) = 23.5^{\circ}\text{C}$. The change is $23.5 - 18.5 = 5^{\circ}\text{C}$. Comparing only the division counts (3 versus 1) ignores that the readings sit above different numbered marks, two degrees apart. The 4-degree gap between the 18 and 22 marks is not the actual change, since the two readings are not on the marks themselves. And treating divisions as tenths gives the wrong readings (18.1 and 23.1); here each division is half a degree, not a tenth, so the values are 18.5 and 23.5.

60. (A) Most of the distance between the Sun and Earth is empty space with no matter, and conduction needs touching particles, so it cannot work across a vacuum — the heat must travel by radiation, which needs no medium. There is no continuous air bridge to the Sun, so conduction through air is impossible. Convection needs a flowing fluid, but space is a vacuum with no fluid to flow, so that fails too. And it is not conduction through the upper air either — the vast vacuum between rules out conduction entirely, leaving radiation as the only mode.